MAIS PAC

IPhO 04.03: Calculus · Lecture + Problems

Calculus

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. Calculus is a physical language

Newton invented the calculus to write down mechanics, and the two ideas it contains are the two ideas mechanics needs:

- A **derivative** is an instantaneous rate of change — velocity from position, acceleration from velocity, current from charge, force from potential energy.
- An **integral** accumulates a rate — distance from velocity, charge from current, work from force, impulse from force.

Everything else in the topic is technique. The physics is in choosing which operation to apply to which quantity, and in recognising when the derivative of a graph is the interesting thing.

2. Differentiation

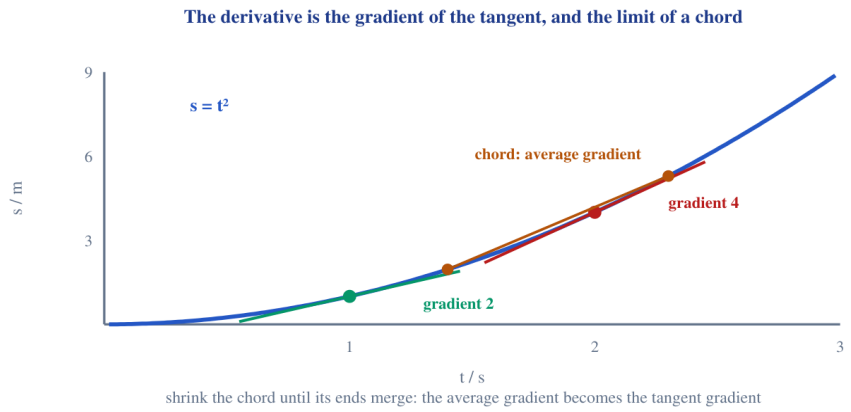


Figure 1. The derivative is local information: it says how fast the quantity is changing at one instant, not on average. That is exactly what Newton's second law needs.

The derivative is defined as a limit,

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}, \tag{1}$$

which for a graph is the gradient of the tangent, and for a physical quantity is the rate of change at an instant.

The rules you actually need:

function	derivative	physics example
x^n	nx^{n-1}	any power law
$\sin x$	$\cos x$	SHM velocity
$\cos x$	$-\sin x$	SHM acceleration
e^{kx}	ke^{kx}	growth and decay
$\ln x$	$1/x$	entropy, potential of a line charge
uv	$u'v + uv'$	product rule
u/v	$(u'v-uv')/v^2$	quotient rule
$f(g(x))$	$f'(g),g'(x)$	chain rule

Two derived facts do most of the work in physics:

- **The chain rule**, which converts a rate in one variable into a rate in another: $\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$.
- **The product rule**, which appears whenever a quantity is a product of two changing factors, such as kinetic energy $\frac{1}{2}mv^2$ or the volume of a balloon.

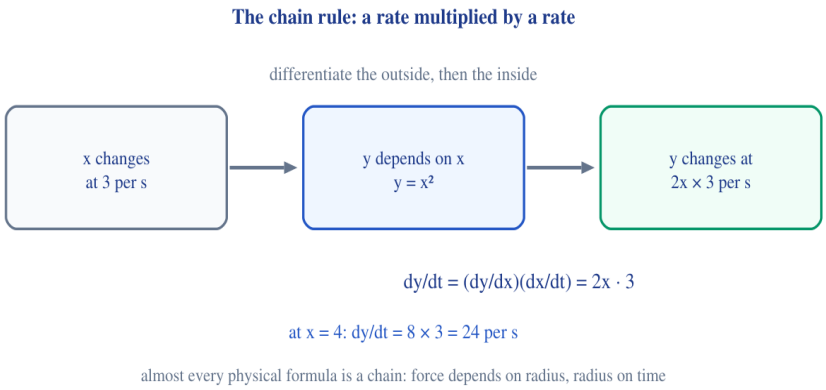


Figure 4. The chain rule is the workhorse of related-rates problems: a balloon inflating, a shadow lengthening, a satellite losing altitude. Write the chain, then substitute.

A useful consequence of the chain rule is the trick of differentiating an implicit relation to turn a constraint into a rate equation. Differentiating $x^2+y^2 = r^2$ gives $2x\,x + 2y\,y = 0$, which is a statement about motion along a circle.

3. Integration

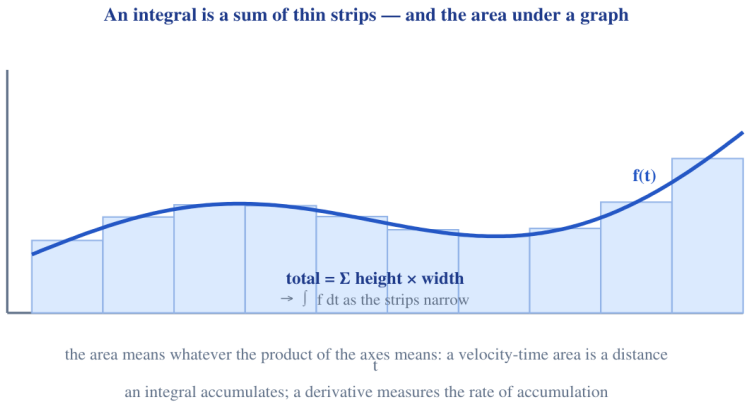


Figure 2. Splitting a quantity into small pieces and adding them is the physical definition of an integral. The calculus is only a way of taking the limit exactly.

Integration is the inverse of differentiation, and a definite integral is the area under a curve. The rule that makes it usable is

$$\int x^n\,dx = (x^{(n+1)})/(n+1) + C\ (n \neq -1),\ \int (dx)/(x) = \ln|x| + C. \tag{2}$$

In physics the limits are almost always supplied by the problem: from the moment the switch closes to the moment you care about, from one position to another. A definite integral has a physical meaning precisely because its limits do.

Three applications are worth knowing by pattern:

- **Work:** $W = \int F\,dx$. For a spring, $\int_0^x kx'\,dx' = \frac{1}{2}kx^2$ — the factor $\frac{1}{2}$ comes from the integration, not from memory.
- **Impulse:** $J = \int F\,dt$. The area under a force–time graph is the change in momentum, which is why airbag graphs are always shaded.

- **Average value:** f = (1)/(b-a)∫_a^b f,dx. The average of sin ^2 over a full cycle is (1)/(2), the result behind rms values.

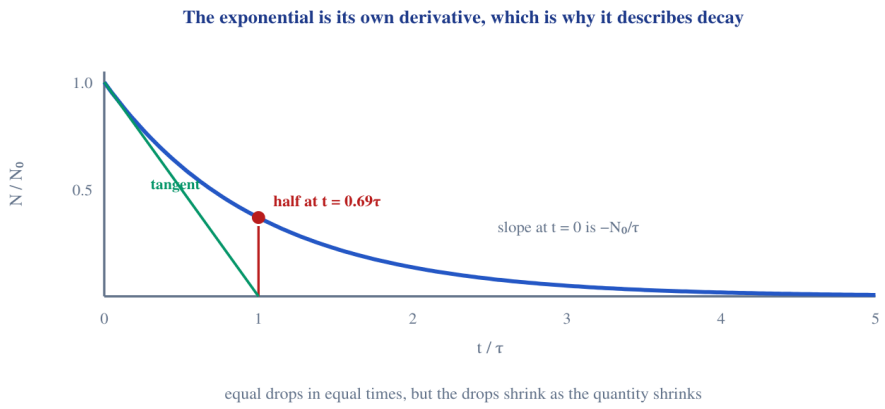


Figure 3. Anything whose rate of loss is proportional to what remains follows this curve: radioactive nuclei, a cooling body, a discharging capacitor, a drug in the blood.

4. Exponential growth and decay

If the rate of change of a quantity is proportional to the quantity itself, (dN)/(dt) = -kN, then N = N_0e^(-kt). This is the only function whose derivative is a multiple of itself, and it is why the same curve appears for radioactive decay, cooling, capacitor discharge and absorption of radiation.

The parameters have standard readings:

- The **time constant** τ = 1/k is the time in which the quantity falls by a factor e.
- The **half-life** t_(1/2) = ln 2/k ≈ 0.693τ is the time in which it halves.
- The initial gradient is -N_0k, and the tangent at t=0 reaches zero at t = τ — a quick graphical way to read a time constant.

5. Taylor series and small quantities

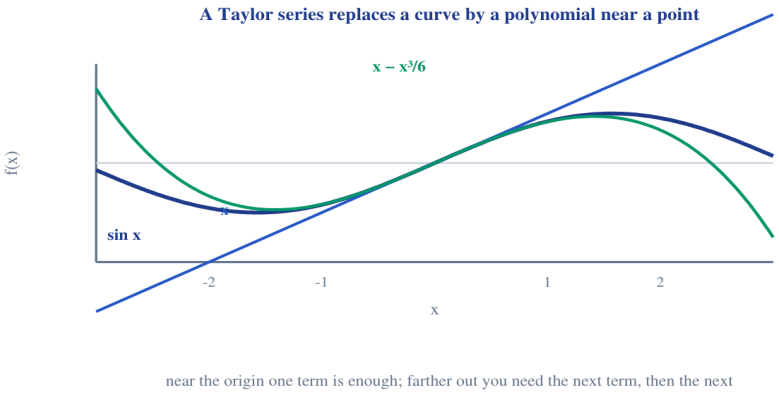


Figure 5. A Taylor series is not an approximation to be ashamed of — it is how small-angle motion, the linear regime of a spring, and almost every "small quantity" argument in physics are justified.

Near x=0,

$$f(x) \approx f(0) + f'(0)x + (f''(0))/(2)x^2 + \quad (3)$$

which gives the approximations you use all the time:

$$\sin x \approx x,\ \cos x \approx 1-(x^2)/(2),\ e^x \approx 1+x,\ (1+x)^n \approx 1+nx. \quad (4)$$

The last one is the binomial approximation, the single most useful line in olympiad physics: it converts $(1+\varepsilon)^n$ into $1+n\varepsilon$ for small ε, which is how you find the "small correction" the examiner is asking for.

The size of the first neglected term is your error estimate. If you use $\sin\theta \approx \theta$ at $\theta = 0.2$, the error is about $\theta^3/6 = 1.3\times10^{-3}$, i.e. 0.7% of the value — a number you can quote.

Interactive experiment: Displacement, gradient and area — open the online handout to use it.

Calculus in a physics answer. Never write a derivative or integral without saying what it means physically: "the gradient of a tangent" or "the area under the graph" earns method marks, and "differentiate" alone does not. When the problem asks for a rate, ask which quantity is changing with respect to which variable, then set up the chain. When it asks for a total, ask which small contribution is being added up.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Rates from a position function. A particle moves with $x = 2t^3 - 9t^2 + 12t$ metres. (a) Find its velocity and acceleration as functions of time. (b) Find the times when it is instantaneously at rest. (c) Determine the displacement over the first 3 {s}.

Worked solution →

[1] Problem 2 — Area under a graph. A car's velocity increases linearly from 0 to 20 {m,s^(-1)} in 10 {s}, stays constant for 20 {s}, then brakes uniformly to rest in 5 {s}. (a) Draw the velocity-time graph. (b) Find the total distance. (c) Find the average speed over the whole motion.

Worked solution →

[2] Problem 3 — Work from a variable force. A spring obeys F = -kx with k = 200 {N,m^(-1)}. (a) Show by integration that the work done to stretch it to x is (1)/(2)kx². (b) Find the work for x = 0.10 {m}. (c) Explain why the force–extension graph gives the work as its area, and why the factor (1)/(2) appears.

Worked solution →

[2] Problem 4 — A related-rates problem. A spherical balloon is inflated at 50 {cm³,s^(-1)}. (a) Write the volume as a function of radius. (b) Differentiate to relate dV/dt and dr/dt. (c) Find the rate of increase of radius when r = 10 {cm}, and when r = 20 {cm}.

Worked solution →

[2] Problem 5 — Simple harmonic motion by differentiation. A mass on a spring has x = Acos ω t. (a) Differentiate twice to show that a = -ω²x. (b) Identify ω for a spring of constant k and mass m. (c) Show that the velocity is a maximum where the displacement is zero.

Worked solution →

[2] Problem 6 — The half-life from an exponential. A sample decays as N = N_0e^(-kt) with k = 0.0693 {s^(-1)}. (a) Find the half-life. (b) Find the fraction remaining after 30 {s}. (c) Find the initial activity if N_0 = 2.0×10^(20) nuclei, and compare with the activity after 30 {s}.

Worked solution →

[2] Problem 7 — The binomial approximation. The period of a pendulum of length L at large amplitude is T = 2π√(L/g)(1+(θ_0²)/(16)). (a) Show the fractional change in period for θ_0 = 10°. (b) Compare the exact T/T_0 with the first-order approximation for θ_0 = 30° and comment. (c) State when the approximation is unsafe.

Worked solution →

[3] Problem 8 — Integrating a force that grows with time. A rocket of mass m is pushed by a force F = F_0(1-t/T) from t = 0 to t = T. (a) Find the impulse delivered. (b) Find the final velocity, starting from rest. (c) Show that the final velocity equals the impulse divided by the mass, and check the units.

Worked solution →

[3] Problem 9 — An RC discharge. A capacitor C discharges through a resistor R so that V = V_0e^(-t/RC). (a) Find the current as a function of time by differentiating. (b) Show that the energy dissipated in the resistor equals the initial stored energy. (c) Explain why the discharge is never complete in finite time.

Worked solution →

[3] Problem 10 — Terminal speed with a drag force. A falling sphere of mass m experiences drag F = -bv. (a) Write the equation of motion and solve for v(t). (b) Show that the acceleration decreases to zero. (c) Find the distance fallen in time t by integrating, and show it approaches v_({term})t for large t.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

Paul's Online Math Notes

Calculus, series and approximation practice at exactly the level the syllabus expects.